

# Building an AI Model for Bird Voice Identification

Birds are beautiful and serve the environment in varied ways. A well-developed AI model can detect and classify bird sounds from large volumes of acoustic data to help monitor biodiversity and support global conservation efforts.

**B**irds contribute to environmental sustainability through various ecological functions. They pollinate plants, disperse seeds, control pests, scavenge, and engineer ecosystems. Birds serve as an indicator species, providing insights into environmental health. They offer cultural and economic benefits, and their droppings enrich soil fertility. Protecting bird

populations and habitats is vital for maintaining these ecological roles and promoting a balanced environment.

The ornithology community is collecting many petabytes of acoustic data every year. This data helps to detect and classify bird sounds, as well as monitor the status and trends of biodiversity in ecosystems. It also helps to identify bird populations and enhance conservation efforts.

## Why bird voice recognition

Bird voice recognition is valuable for ornithology and conservation, especially in biodiverse areas. Here are a few benefits.

- *Bird identification:* Non-invasive and effective for detecting cryptic species.
- *Bird population monitoring:* Helps estimate population sizes and monitor changes over time.
- *Behavioural studies:* Provides insights into communication, social structures, and stress indicators.
- *Habitat usage:* Maps habitat preferences and assesses environmental changes.
- *Migrant monitoring:* Tracks migratory patterns and indicates climate change impacts.
- *Conservation efforts:* Aids in biodiversity assessments and early detection of invasive species.

## Challenges of bird voice identification

Due to the complexity of bird vocalisations and ambient conditions, bird voice identification has numerous problems. Some of the obstacles are:

- *Intra-species variation:* Age, sex, health, and geographic location affect a bird's call within its species.
- *Inter-species similarities:* Differentiating one species from another when they have remarkably similar calls is a challenge.
- *Environmental noise:* Wind, water, other animals, and human activity



Image source: <https://www.freepik.com/>